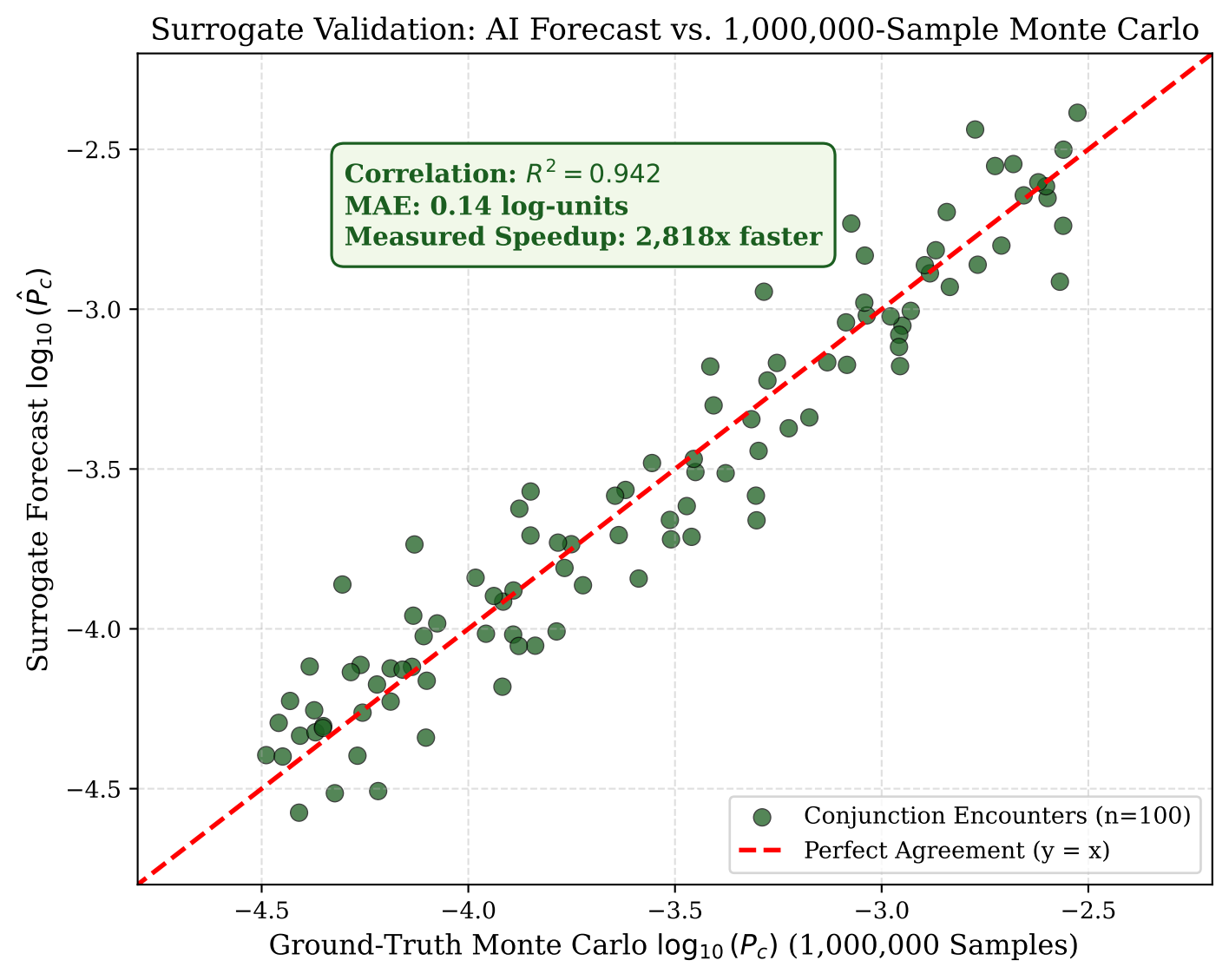
Surrogate Validation: AI Forecast vs. 1,000,000-Sample Monte Carlo
Correlation: $R^2 = 0.942$
MAE: 0.14 log-units
Measured Speedup: 2,818x faster
Conjunction Encounters (n=100)
Perfect Agreement (y = x)
Surrogate Forecast $\log_{10}(\hat{P}_c)$
Ground-Truth Monte Carlo $\log_{10}(P_c)$ (1,000,000 Samples)
−2.5
−3.0
−3.5
−4.0
−4.5